DISSECTING CHRONOS: SPARSE AUTOENCODERS REVEAL CAUSAL FEATURE HIERARCHIES IN TIME SERIES FOUNDATION MODELS

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1. MOTIVATION / PROBLEM

- Time series foundation models are increasingly used in high-stakes forecasting, but their learned representations remain poorly understood.
- Mechanistic interpretability has advanced rapidly in language models with SAEs; TSFMs have not had the same causal validation.
- This work asks whether learned SAE features are actual intervention handles for Chronos forecasts, not just attractive labels.

Model: Chronos-T5-Large, 710M parameters, T5 encoder-decoder.

Gap: feature labels alone do not prove causal relevance.

Test: zero individual SAE features and measure CRPS damage.

Prior interpretability stops short: saliency, perturbation, counterfactual, and concept methods explain behavior from outside the model; this work intervenes on individual SAE codes inside Chronos.

Poster question: can sparse Chronos features act as causal controls for probabilistic forecasts?

2. CORE CONTRIBUTIONS

1. Train TopK SAEs on Chronos-T5-Large activations at six extraction points.
2. Validate causal relevance with 392 single-feature ablations; every tested feature increases CRPS.
3. Reveal a depth hierarchy: Enc. 5 captures low-level cues, Enc. 11 is the causal bottleneck, and Enc. 23 is the semantic map.

Depth hierarchy in one line

Enc. 5 local frequency cues → **Enc. 11** causal bottleneck max ΔCRPS 38.61 → **Enc. 23** 59.8% labeled semantic map

Core inversion: the semantically richest layer is not the most causally important layer; intervention damage peaks in the mid-encoder.

4. KEY QUANTITATIVE RESULTS

392 / 392 single-feature ablations caused positive ΔCRPS

5.15 mean ΔCRPS at encoder block 11; max = 38.61

3.05 / 0.95 / 26.32 encoder block 5 mean / median / max

3.73 / 2.98 / 11.65 encoder block 23 mean / median / max

2.37 / 2.37 / 2.44 extended block 23 sweep over 200 features

Global labeled features: 8,462 / 49,152 = 17.2%

Reader path: first identify causally damaging features, then compare where semantic labels concentrate across depth.

3. MODEL & METHOD

Chronos-T5-Large
710M params
T5 encoder-decoder
24 encoder + 24 decoder layers
$d_{\text{model}} = 1024$
4096-dim tokenization
Hidden states are the intervention substrate.
General-purpose pretrained TSFM, tested without weight updates.
Role: frozen model under causal edits.

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Forward hooks
Enc. block 5
Enc. block 11
Enc. block 23
Dec. block 11
Dec. block 11 cross-attn
Dec. block 23
Hooks read activations without retraining Chronos.
Six points compare depth and encoder/decoder roles.
Role: extraction map across depth.

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Train TopK SAEs
$d_{\text{SAE}} = 8192$
$k = 64$
50,000 steps
Adam, lr $= 3 \cdot 10^{-4}$
64 active features per encoded state.
49,152 total sparse features across six SAEs.
Role: sparse intervention basis.

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Synthetic diagnostics
Trend
Seasonality
Level shifts
Frequency range
Heteroscedastic noise
Labels are assigned by diagnostic correlation.
Unknown is retained when correlation is weak.
Role: semantic taxonomy.

→

Causal validation
ETT benchmark
Single-feature ablation
ΔCRPS = ablated - original
Positive means worse forecast
Decode, patch, and rescore.
392 single-feature interventions plus progressive sweeps.
Role: causal validity check.

Taxonomy route: correlate SAE activations on synthetic diagnostics with ground-truth temporal properties; below-threshold features remain unknown.

Causal route: zero a single sparse code, decode modified activations, patch them into Chronos, and measure ΔCRPS.

DATA SOURCES

Synthetic diagnostics suite: trends, seasonality, level shifts, frequency sweeps, heteroscedastic noise

ETT benchmark: causal ablation experiments with CRPS degradation

Experimental design: TopK makes sparse codes directly ablatable; six extraction points separate early, middle, and late encoder/decoder behavior; CRPS evaluates probabilistic forecast quality rather than only point error.

5. SINGLE-FEATURE ABLATION SUMMARY

Table 1. Single-feature ablation summary across encoder layers.

Layer	n	Mean	Med.	Max	Std	Max/Med
Enc. block 5	64	3.05	0.95	26.32	5.12	27.7x
Enc. block 11	**64**	**5.15**	**1.26**	**38.61**	**6.95**	**30.5x**
Enc. block 23	64	3.73	2.98	11.65	2.07	3.9x
Enc. block 23†	200	2.37	2.37	2.44	0.03	1.03x

† Extended sweep over 1,024 windows with 8 samples. All values report $\text{CRPS}_{\text{ablated}} - \text{CRPS}_{\text{original}}$.

Main empirical split
Enc. 11 is the heavy-tailed causal layer: mean 5.15, max 38.61, max/median 30.5x.
Enc. 23 is semantically broad but flatter under ablation: 200-feature sweep max/median 1.03x.

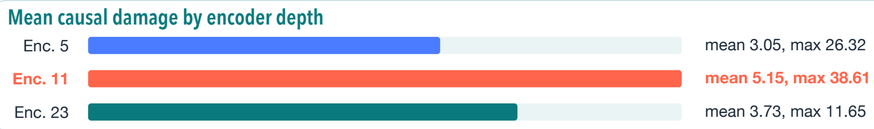
Feature 4616: strongest observed block-11 feature, ΔCRPS = 38.61.

Uniform pruning risk: a small mid-layer feature set can carry disproportionate forecast leverage.

- Block 11 has the highest mean damage and the most extreme single-feature failures.
- Block 23 has the broadest taxonomy, but its extended ablation curve stays almost flat.
- The main empirical split is therefore not "interpretable vs. uninterpretable"; it is semantic breadth vs. causal concentration.

Intervention protocol: zero a single sparse feature code, decode the modified activation, patch it into the Chronos forward pass, then recompute probabilistic forecast quality with ΔCRPS.

Mean causal damage by encoder depth

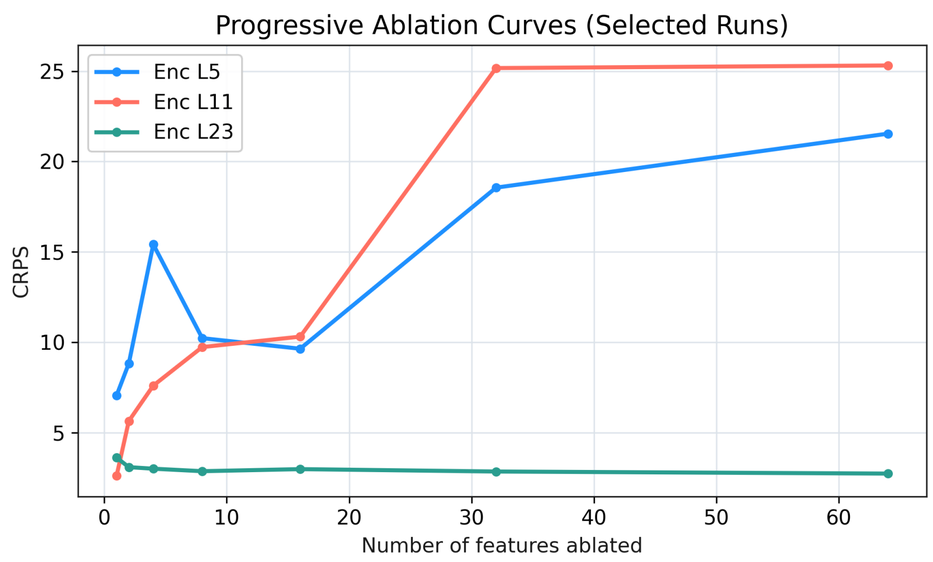


Enc. 5: early causal signal; high max but sparse labels.

Enc. 11: strongest intervention target and heaviest tail.

Enc. 23: broad taxonomy; flatter single-feature damage.

6. PROGRESSIVE ABLATION CURVES



100% positive: all tested single-feature ablations worsen CRPS.

38.61: max single-feature ΔCRPS at encoder block 11.

Mid-encoder: causal bottleneck under progressive ablation.

Enc. 5: $7.05 \rightarrow 21.54$ with only 4.8% active features.

Enc. 11: $2.61 \rightarrow 25.32$ and 23.3% active features.

Enc. 23: $3.62 \rightarrow 2.73$ despite 54.0% active features.

Interpretation: dense semantic coverage in the late encoder does not translate into cumulative damage under progressive ablation.

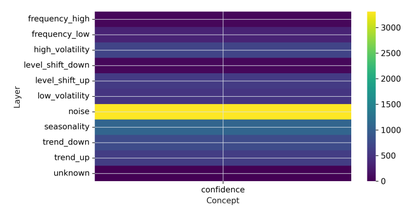
Reading the curves

Block 11 grows from small single-feature penalties to a steep cumulative jump; block 23 stays nearly flat even when many active features are removed.

7. FEATURE TAXONOMY

Table 2. Feature taxonomy counts by extraction point.

Concept	Enc 5	Enc 11	Enc 23	Dec 11	Dec 11_{ca}	Dec 23
Seasonality	12	45	1,439	33	91	64
Level shift ↑	66	1,024	1,097	28	37	28
Level shift ↓	46	–	210	14	26	60
Trend ↑	48	220	228	13	12	37
Trend ↓	11	16	89	–	13	32
Freq. high	97	91	668	16	–	24
Freq. low	10	–	456	83	82	93
High volatility	68	268	260	–	13	14
Low volatility	14	28	138	33	43	94
Noise	32	413	315	24	17	–
Labeled (%)	**4.9**	**25.8**	**59.8**	**3.1**	**4.2**	**5.5**



Late encoder concentration: Enc. block 23 reaches 59.8% labeled features.

Decoder sparsity: decoder labeled fractions remain 3.1% to 5.5%.

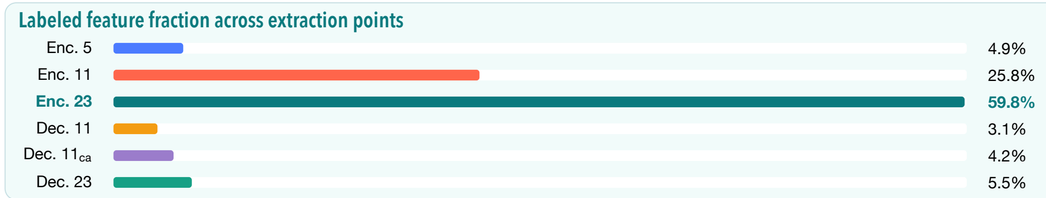
Across all 49,152 SAE features, 8,462 receive non-unknown labels (17.2%). Labels are heuristic correlation-based assignments on synthetic diagnostics; Dec 11_{ca} denotes cross-attention output at decoder block 11.

Early: Enc. 5 is causal but sparsely labeled at 4.9%.

Middle: Enc. 11 has strong level-shift/noise counts and high causal damage.

Late: Enc. 23 peaks for seasonality and upward level shifts.

Labeled feature fraction across extraction points



Largest cells: seasonality 1,439 and level shift ↑ 1,097 at Enc. 23.

Block 11 role: fewer labels than Enc. 23, but stronger causal damage.

Unknown is retained: unlabeled features are not forced into weak concepts.

8. MAIN INSIGHTS / DISCUSSION

Depth is functional

Chronos features are not uniformly distributed across depth: early layers show local temporal structure, block 11 concentrates causal leverage, and block 23 carries the broadest semantic vocabulary.

Causality and interpretability diverge

The final encoder contains the richest taxonomy, but the mid-encoder is where ablations do the most damage. Semantic richness alone is not the same as causal importance.

Operational implication

For TSFM editing, pruning, or safety work, the strongest first target is not the most label-rich layer; it is the layer whose feature interventions reliably degrade forecasts.

Reusable circuit question: do the same features remain causal across datasets?

Scaling question: do larger SAEs expose circuit-level structure?

Evidence ladder

1. **Label features:** synthetic diagnostics assign trend, seasonality, shift, frequency, volatility, and noise concepts.
2. **Intervene:** ablate one feature at a time and measure the forecast penalty with ΔCRPS.
3. **Separate roles:** Enc. 11 is the causal bottleneck; Enc. 23 is the semantic map.

Practical readout: use block 11 when the goal is controlled model editing or forecast stress testing; use block 23 when the goal is semantic auditing of what Chronos represents.

Target selection guide

Intervene: start with Enc. 11 features and inspect heavy-tail failures.

Audit: use Enc. 23 to map the richest temporal vocabulary.

Screen: use Enc. 5 to catch early low-level temporal features.

9. LIMITATIONS & FUTURE WORK

Limitations

- Taxonomy labels are heuristic; 82.8% of all features remain unlabeled.
- Causal experiments are limited to Chronos-T5-Large and the ETT benchmark.
- Single-feature ablations expose local causal handles, not complete circuits.

Next experiments

- Train larger SAEs and compare TopK choices across layer depths.
- Replace heuristic labels with stronger labelers and human-audited concepts.
- Test cross-architecture stability and trace circuits behind block-11 features.

Bottom line: Chronos appears to form a real depth hierarchy: low-level temporal cues emerge early, the mid-encoder carries the sharpest causal leverage, and the late encoder compresses the broadest semantic vocabulary.

Do not overclaim

- The result supports causal feature handles in the tested Chronos-T5-Large setting.
- It does not yet establish universal circuit structure across all TSFMs or datasets.

Replication target: same ablation protocol across more forecasting datasets.

Mechanistic target: trace block-11 features into reusable circuits.

Conference takeaway: sparse feature interventions convert TSFM interpretability from a descriptive map into a causal experiment.

Main run: 256 ETT windows, 4 forecast samples.

Extended run: 1,024 windows, 8 samples, 200 features.

Relation to prior work

Post-hoc TS methods: saliency, perturbations, counterfactuals, and concepts explain forecasts externally.

SAE labeling alone: maps feature meaning but does not prove the feature matters to the forecast.

This poster: ablates one SAE code at a time and measures forecast degradation.